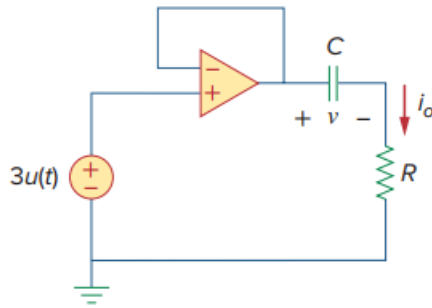


Problem

Find i_o in the op amp circuit in Fig. 7.137. Assume that $v(0) = -2$ V, $R = 10$ k Ω , and $C = 10$ μ F.

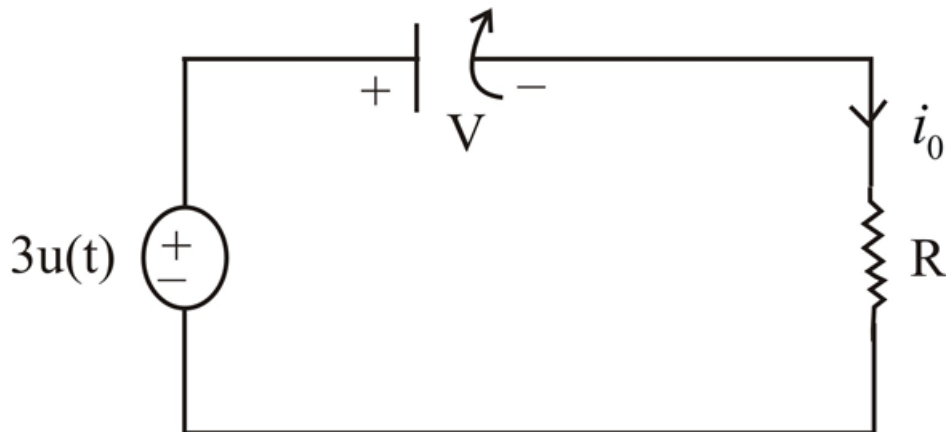
Fig. 7.137:



Step-by-step solution

Step 1 of 2

The op-amp acts as an emitter follower so that the Thevenin's equivalent circuit is shown below.



Step 2 of 2

Hence,

$$V(t) = V(\infty) + [V(0) - V(\infty)]e^{-\frac{t}{\tau}}$$

$$V(0) = -2 \text{ V.}$$

$$V(\infty) = 3 \text{ V}$$

$$\tau = RC = (10 \times 10^3)(10 \times 10^{-6}) = 0.1$$

$$V(t) = 3 + (-2 - 3)e^{-10t}$$

$$= 3 - 5e^{-10t}$$

$$i_o(t) = C \frac{dV}{dt} = (10 \times 10^{-6})(-5)(-10)e^{-10t}$$

$$i_o(t) = 0.5 e^{-10t} \text{ mA}, \quad t > 0$$